

Dose of Reality

Overlap Deceives, Exceedance Is Real

Ingredient Overlap and Safe Upper Limit Exceedance in U.S. Dietary Supplement Use: An NHANES 1999-2023 Analysis

Workintech Data Science Bootcamp — Capstone Project #2

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46,388 person-cycle observations · 10 NHANES cycles (1999-2023) · BigQuery + dbt + Python · svy (design-based statistics)

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Abstract

This study examines dietary supplement usage behavior in the United States using NHANES (National Health and Nutrition Examination Survey) data spanning 1999-2023 — 24 years and 10 survey cycles, analyzed under NHANES's official survey-weighting rules. The project originated from a personal observation: people who take more than one supplement may unknowingly consume the same vitamin or mineral from multiple products, and this overlap may push their total intake past the official Tolerable Upper Intake Level (UL).

Across the six nutrients examined (Vitamin D, Iron, Magnesium, Niacin, Zinc, Vitamin A), overlap and UL exceedance were found to differ significantly by gender, age group, and ethnicity (10 of 11 tests at $p < 0.0001$). 64.2% of exceedance cases ($p < 0.0001$) were found to stem from overlap rather than a single product's own dosage. The COVID-19 pandemic period (2021-2023) was found to have driven overall usage 20.9 percentage points below what the pre-pandemic trend would have predicted, while simultaneously intensifying multi-product use among those who continued using supplements.

The findings suggest that public health messaging should combine both a 'watch for overlap across products' message and a 'read the single-product dosage label' message — and that these messages may need to be tailored by nutrient and by demographic group.

1. Introduction and Motivation

This project began with the observation that using multiple dietary supplements may carry a risk that users don't notice: two separate products from different brands may contain the same vitamin or mineral, and the combined intake may exceed the scientifically established safe upper limit (UL). This study turns that personal question into a population-level behavior and risk analysis.

Research Questions: (1) What is the level of dietary supplement use and multi-product overlap in the U.S. population, and how has it changed over time? (2) To what extent does overlap translate into actual dose exceedance? (3) Do these behavior and risk patterns differ significantly by gender, age, and ethnicity? (4) What was the effect of the COVID-19 pandemic on these patterns?

This project deliberately excludes drug-supplement interactions and individual clinical dosage advice; the goal is to produce a population-level picture of behavior and trends using open-source data.

2. Data and Methods

2.1 Data Source

The analysis uses the CDC's National Health and Nutrition Examination Survey (NHANES) dataset, covering 10 survey cycles from 1999 to 2023. The 1999-2002 period was combined into a single 4-year block per CDC's official recommendation (using the special WTINT4YR weight); 2017-2020 is a special cycle extended to 3.2 years due to the pandemic.

2.2 Data Pipeline

Data was processed through a layered ELT architecture using BigQuery and dbt: Raw (raw .xpt files) → Staging (code decoding, canonicalization) → Intermediate (joins, unit standardization) → Marts (analysis-ready summary tables). All code is version-controlled on GitHub.

2.3 The Product ID System Discontinuity

Between 2016 and 2017, NHANES changed its product identification numbering system (DSDSUPID → DSDPID). A cross-mapping column within the current reference table (DSPI) was used to achieve a 90-98% match rate across all periods.

2.4 Weighting

In accordance with NHANES's official analytic guidelines, interview weights were used, since the Dietary Supplement Questionnaire is administered as part of the household interview. When pooling multiple cycles, a duration-weighted formula consistent with NHANES's official 'combining cycles' rule was applied: $\text{combined_weight} = (\text{cycle duration} / \text{total duration}) \times \text{the cycle's own weight}$.

2.5 UL Comparison and Statistical Method

Official UL values from the Institute of Medicine / National Academy of Sciences (IOM/NAS) were used as reference. Six ingredients exceeding a 2% population exceedance threshold in preliminary analysis were included: Vitamin D, Iron, Magnesium, Niacin, Zinc, and Vitamin A. Statistical tests used design-based methods that account for NHANES's complex survey design (Python's svy package: t-tests, Kruskal-Wallis rank tests, one-sample proportion tests).

3. Overall Usage Trend

Overall usage rose steadily from 1999 to 2020 (51.7% → 57.7%). In 2021-2023, usage dropped to 34.3%, while multi-product use among users rose from 55.4% to 61.7% — fewer people were using supplements, but those who did were using them far more intensively.

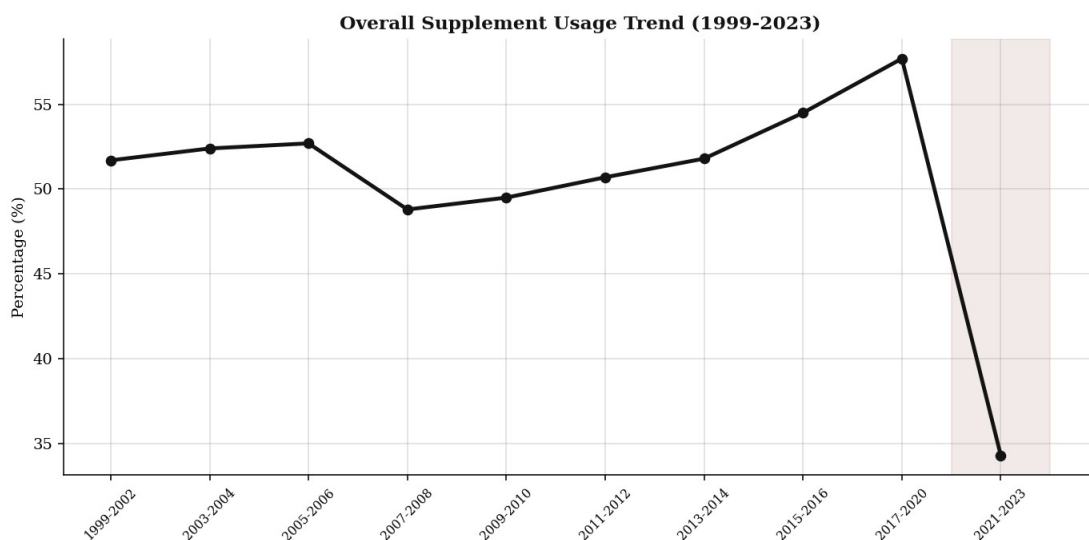


Figure 1. Overall supplement usage trend, 1999-2023.

4. Overlap and UL Exceedance Trend (6 Ingredients)

Vitamin D shows a continuous, accelerating trend — from near-zero UL exceedance in 1999-2006 (0.0-0.1%) to 27.5% in 2021-2023. Zinc, by contrast, stayed flat (3-4%) for over two decades and only spiked to 9.4% in the most recent cycle — a new behavior rather than an accelerating one.

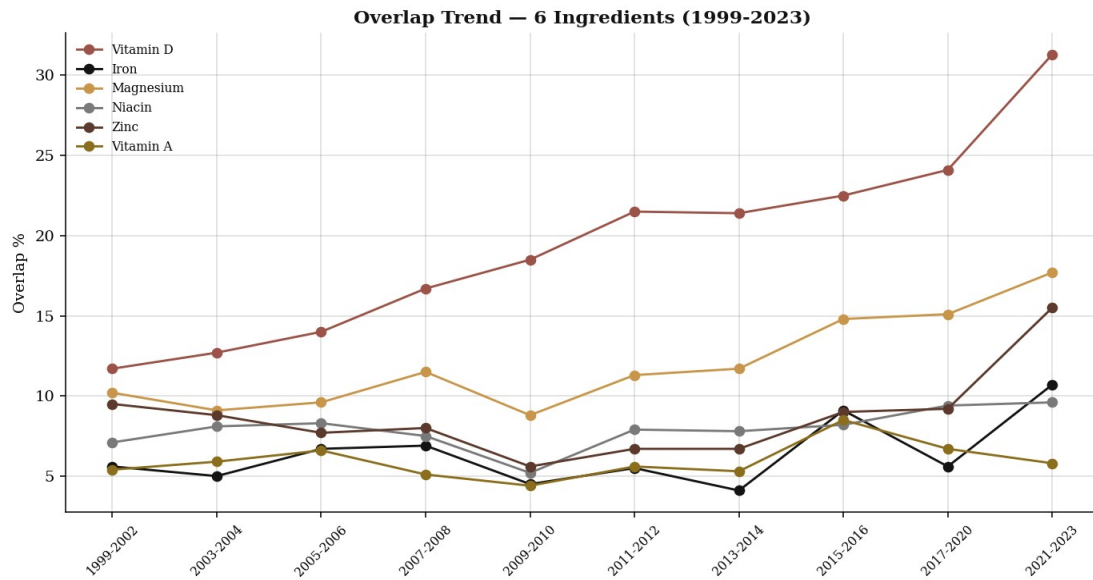


Figure 2. Overlap trend, 6 ingredients (1999-2023).

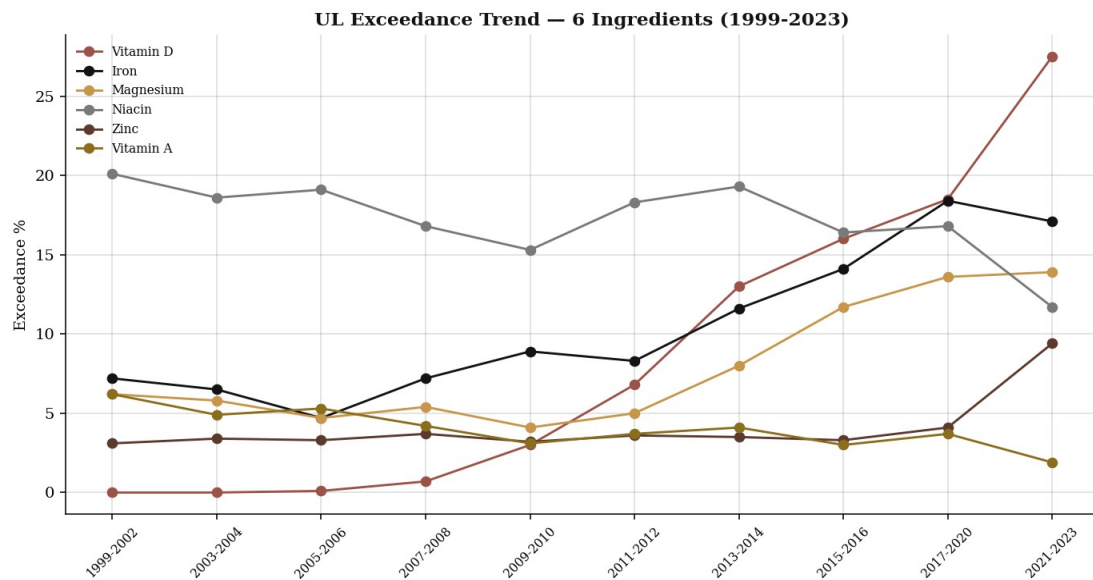


Figure 3. UL exceedance trend, 6 ingredients (1999-2023).

Niacin and Vitamin A, unlike the other four ingredients, show a general decline over 24 years — a trend independent of the pandemic.

5. Source of Exceedance: Overlap or Single Product?

Each ingredient carries its own risk mechanism: for Vitamin D, risk comes almost entirely from overlap; for Iron, risk comes largely from single-product dosage and peaked in 2017-2020 before receding. Niacin's 24-year decline is entirely driven by its single-product component.

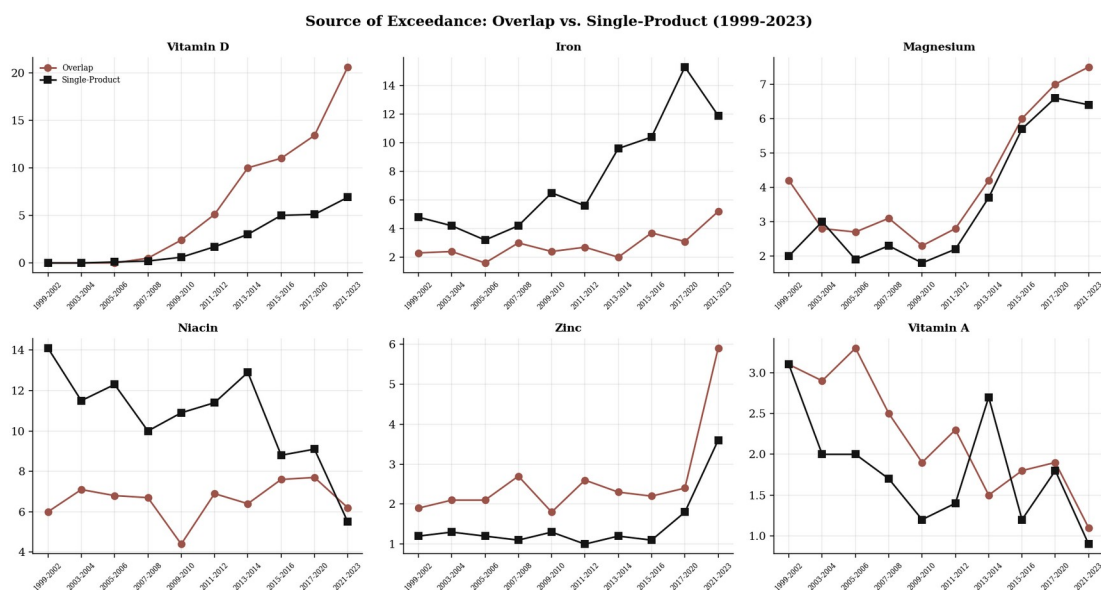


Figure 4. Source of exceedance — overlap vs. single-product, 6 ingredients (1999-2023).

6. Demographic Breakdowns

6.1 Gender

Women have consistently shown higher usage, overlap, and UL exceedance rates than men across 24 years. The gender gap in UL exceedance has widened systematically since 2017; its source is women's rising multi-product (overlap) behavior.

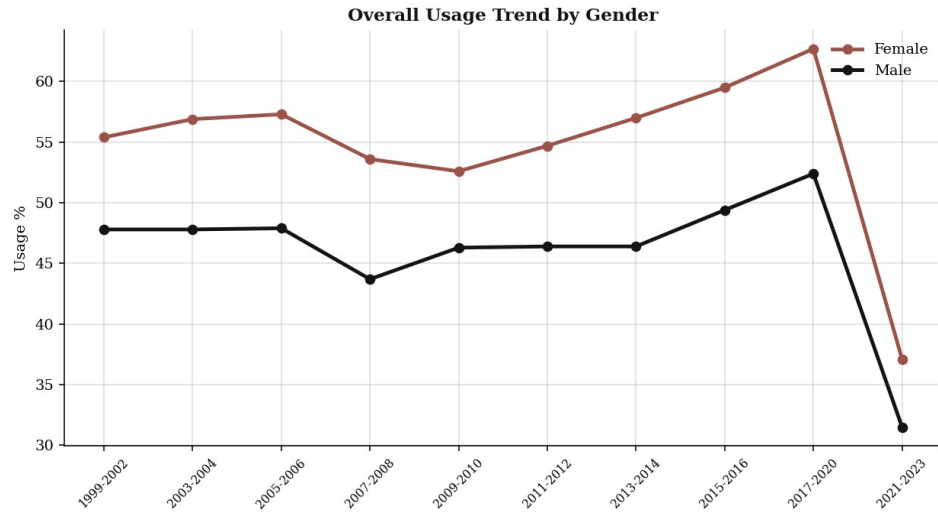


Figure 5. Overall usage trend by gender.

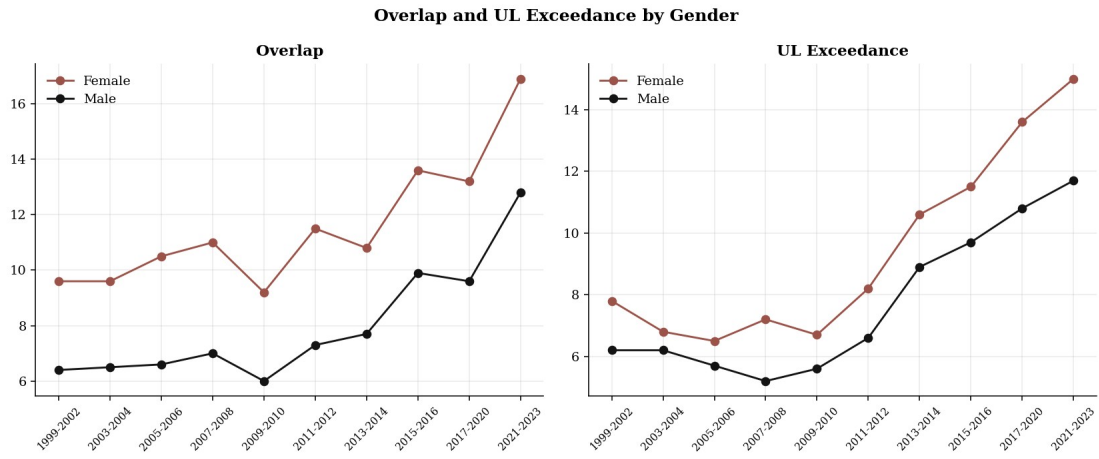


Figure 6. Overlap and UL exceedance by gender.

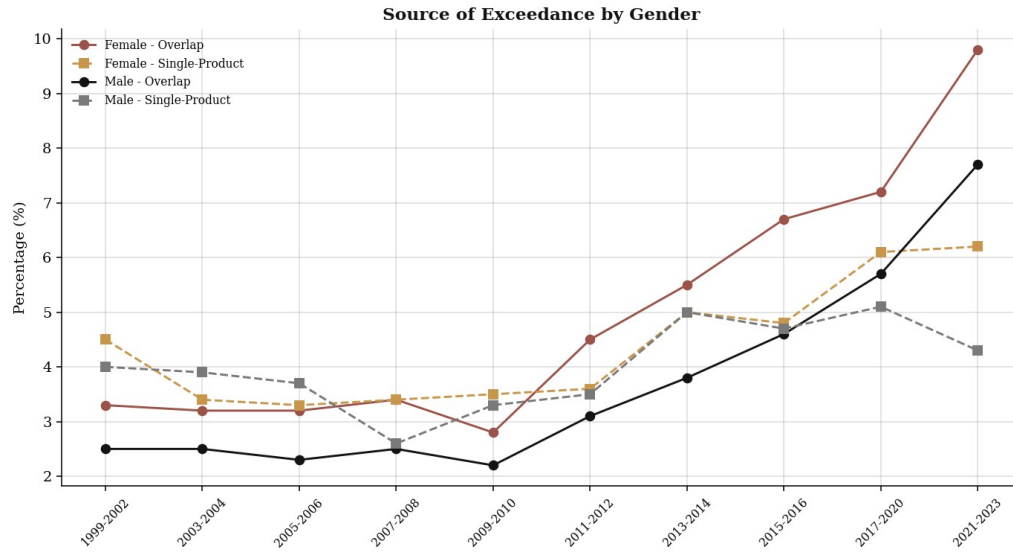


Figure 7. Source of exceedance by gender.

6.2 Age Group

Usage, overlap, and exceedance rates rise monotonically with age (65+ is always highest). This pattern was not yet fully established in 1999-2010 and became clear from 2011 onward.

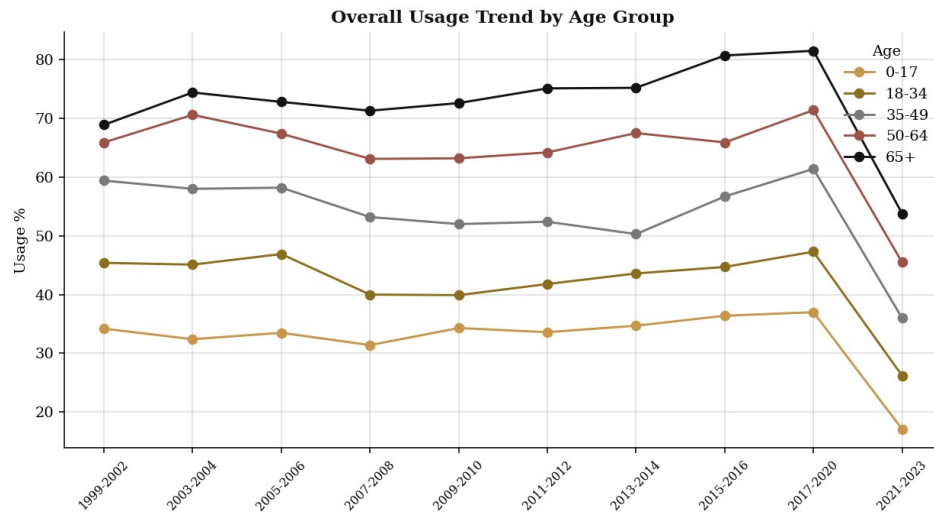


Figure 8. Overall usage trend by age group.

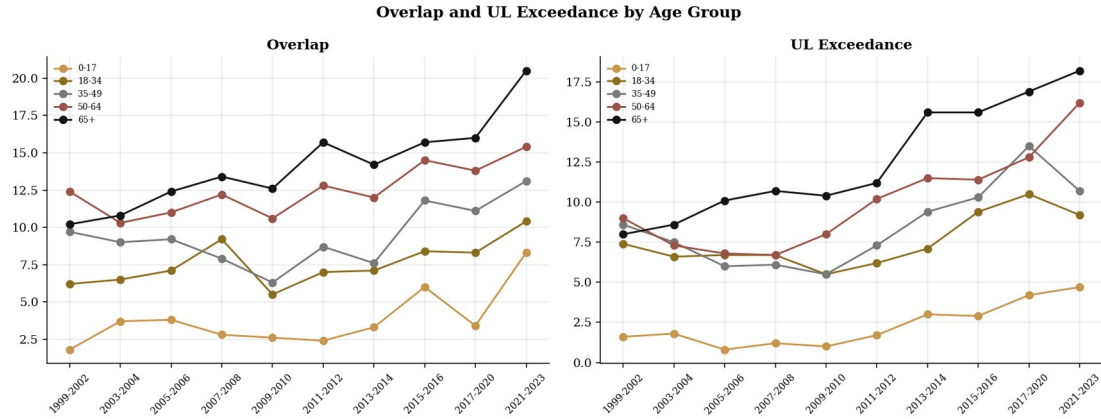


Figure 9. Overlap and UL exceedance by age group.

6.3 Ethnicity

The Non-Hispanic White group has consistently led in overlap. In UL exceedance, however, the Non-Hispanic Black group converged starting in 2015-2016 and overtook the White group in 2021-2023.

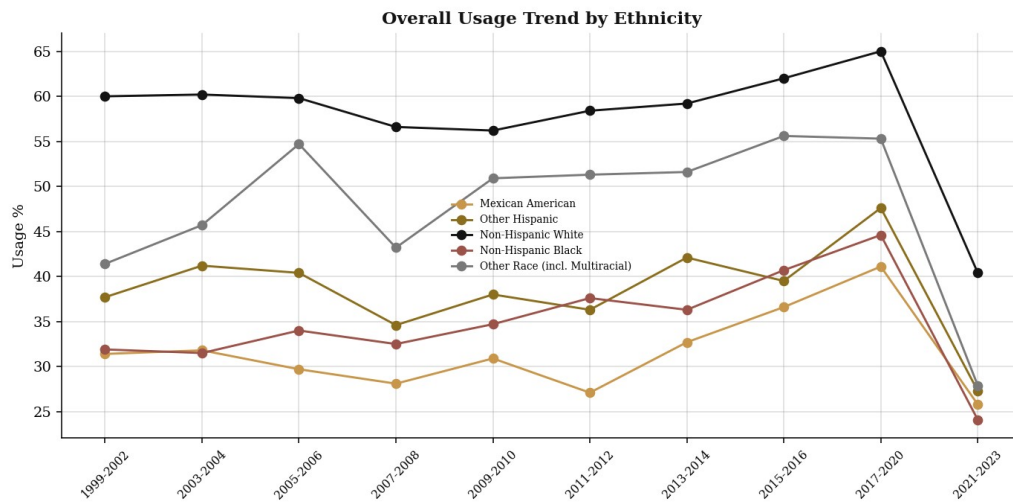


Figure 10. Overall usage trend by ethnicity.

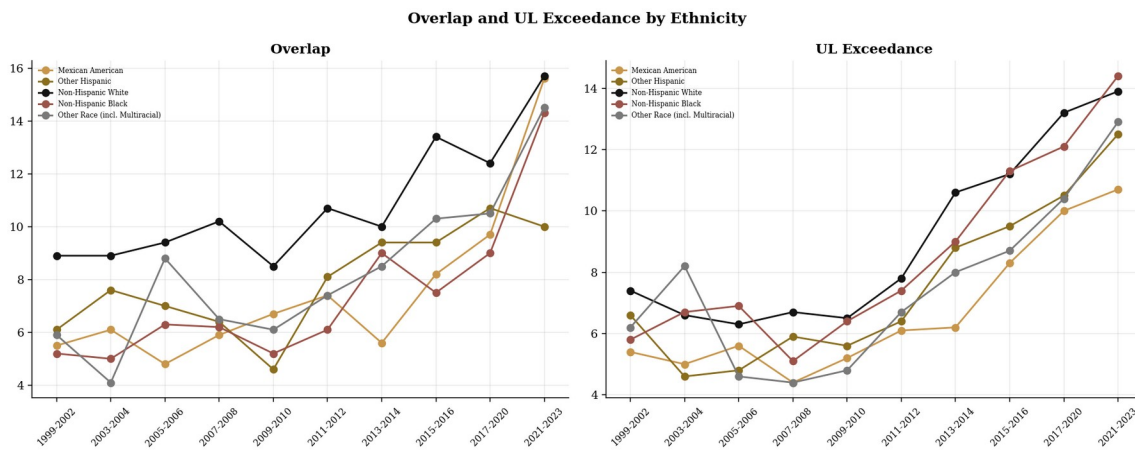


Figure 11. Overlap and UL exceedance by ethnicity.

7. Counterfactual Analysis

Rather than forecasting the future directly, a model built on the 1999-2020 pre-pandemic trend produced a 'what if the trend had continued' projection for 2021-2023, which was then compared against the actual observed value.

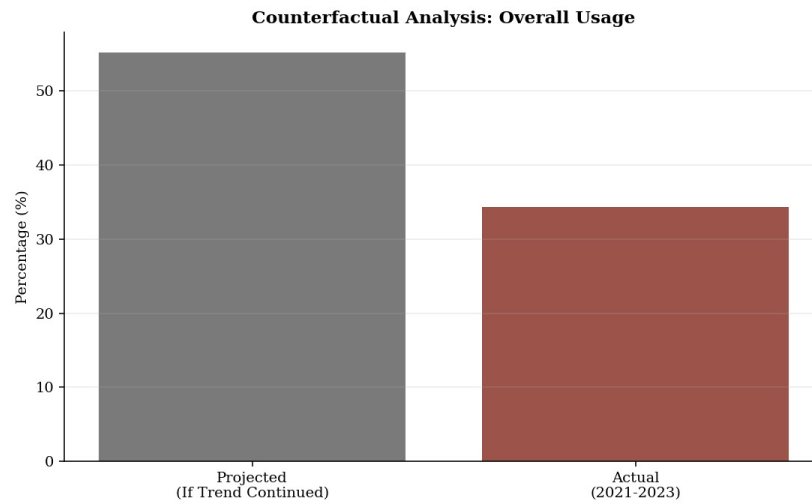


Figure 12. Overall usage — projected vs. actual.

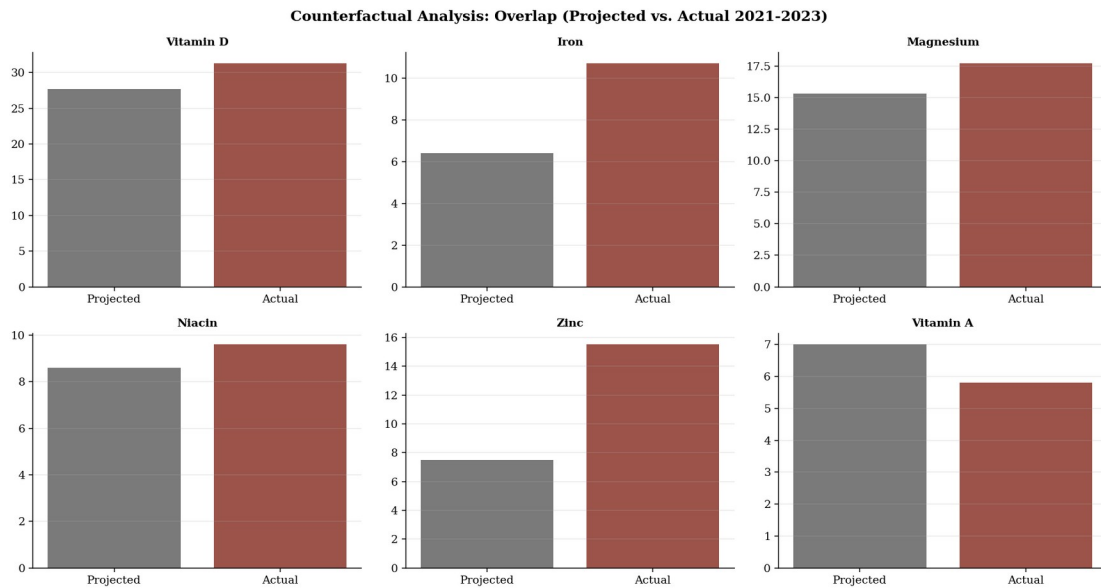


Figure 13. Overlap — projected vs. actual, 6 ingredients.

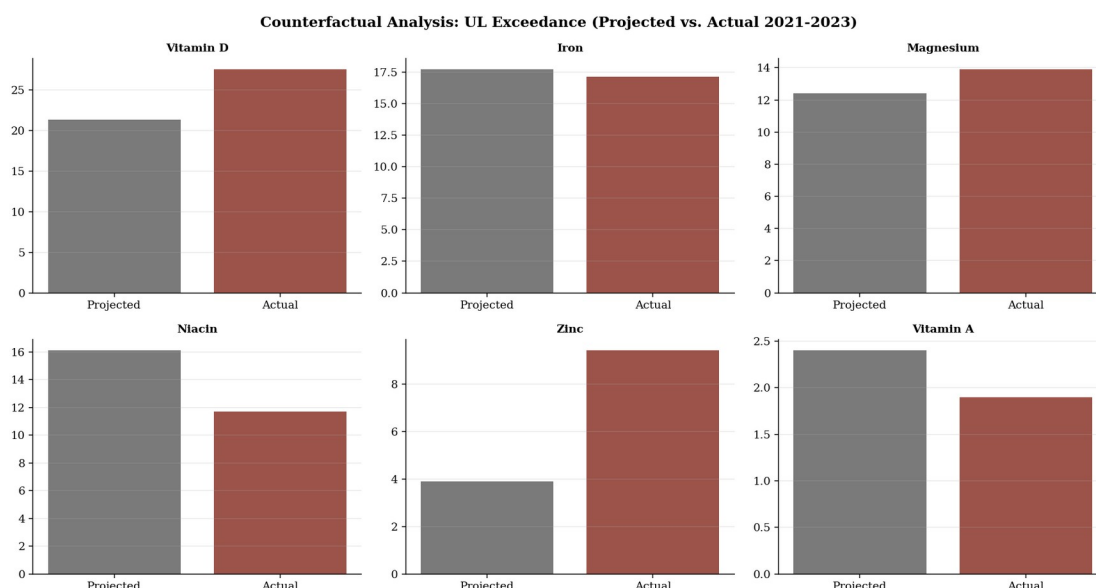


Figure 14. UL exceedance — projected vs. actual, 6 ingredients.

8. Statistical Significance Tests

Design-based tests, accounting for NHANES's complex survey design, were applied throughout.

Test	Period	Statistic	p-value
Gender × Overlap	24 years	$t = -13.76$	< 0.0001
Gender × UL Exceedance	24 years	$t = -9.25$	< 0.0001
Age Group × Overlap	24 years	$F = 584.79$	< 0.0001
Age Group × UL Exceedance	24 years	$F = 531.62$	< 0.0001
Ethnicity × Overlap	24 years	$F = 114.37$	< 0.0001
Ethnicity × UL Exceedance	24 years	$F = 12.69$	< 0.0001
Overlapping vs. Single-Product (against 50%)	24 years	$t = 21.37$	< 0.0001
Iron × Targeted Subgroup (n=15)	2021-2023	$t = 1.33$	0.2049 (not significant)

Seven of eight tests are significant at $p < 0.0001$. The sole exception is a hypothesis about whether Iron exceedance is specific to the perimenopausal period, tested on a very small subgroup ($n=15$) — a result indicating insufficient statistical power, not an invalid hypothesis.

9. Discussion and Limitations

9.1 Methodological Lessons

- An accidental 4x duplication in raw data loading was detected and corrected through cross-validation.
- Early analyses used unweighted statistics before switching to weighted analysis to comply with NHANES's official guidelines. The magnitude of this correction depends on the type of question:

small (<1 point) for conditional rates (e.g., overlap among users), large (~25-29 points) for overall prevalence rates.

- The historical discontinuity in product ID numbering (1999-2016 vs. 2017-2023) was successfully bridged using a cross-mapping column embedded within the source table itself.

9.2 Limitations

- The pandemic effect and a concurrent survey-mode change (in-person to telephone) occurred simultaneously in 2021-2023; their relative contributions cannot be disentangled with this data.
- Large differences observed in small demographic subgroups (n<20) should be interpreted with caution; sample sizes are reported alongside every such finding throughout this project.
- This is an observational cross-sectional/trend analysis; findings are correlational, not causal (e.g., the overlap between the ethnicity × Iron exceedance finding and known anemia prevalence disparities is presented as a hypothesis, not a proven causal link).
- Drug-supplement interactions and individual clinical dosage safety are outside this study's scope.

10. Conclusion

This study shows that dietary supplement use in the U.S. rose between 1999 and 2023, but a substantial share of that trend is driven by multi-product overlap, and that overlap leads to statistically significant safe-upper-limit exceedance. Risk cannot be reduced to a single demographic group or a single mechanism (overlap vs. single-product dosage) — each nutrient carries its own risk profile. The COVID-19 pandemic materially disrupted overall usage behavior while simultaneously concentrating risk within the remaining user base.

11. Closing Remarks

This project began with a personal question and, through an end-to-end data engineering and statistical analysis pipeline built entirely on open data, arrived at population-level findings. Technical challenges encountered along the way — data quality errors, historical ID system discontinuities, weighting decisions — were documented transparently and corrected, demonstrating that not only the findings but the methodological process itself is sound and reproducible. Future work could pursue full statistical interaction models (three-way demographic interactions) and larger targeted subgroup samples.

References

Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS). National Health and Nutrition Examination Survey (NHANES), 1999-2023 Data Files.

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Institute of Medicine (IOM), Food and Nutrition Board. Dietary Reference Intakes (2000, 2003).

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